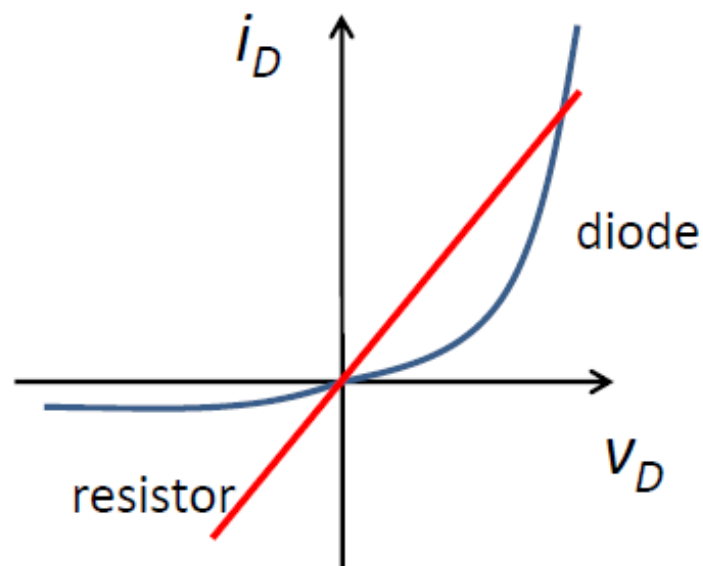
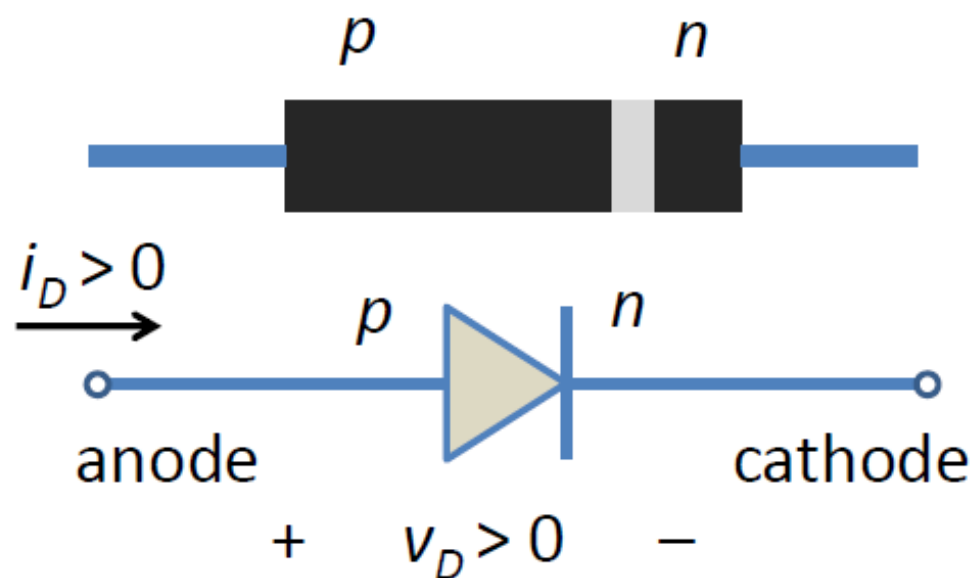


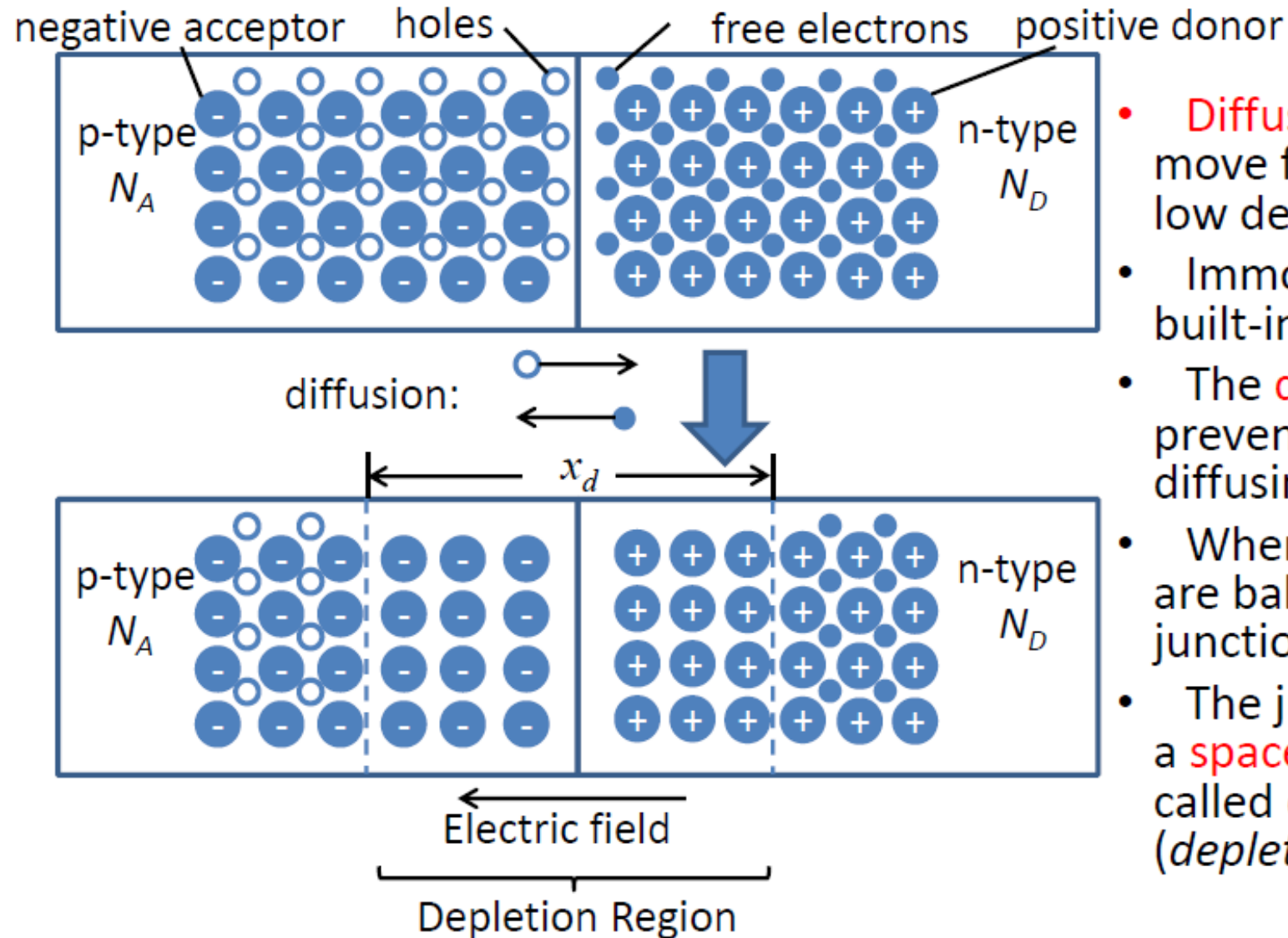
# Diodes

# P-N Junction Diodes



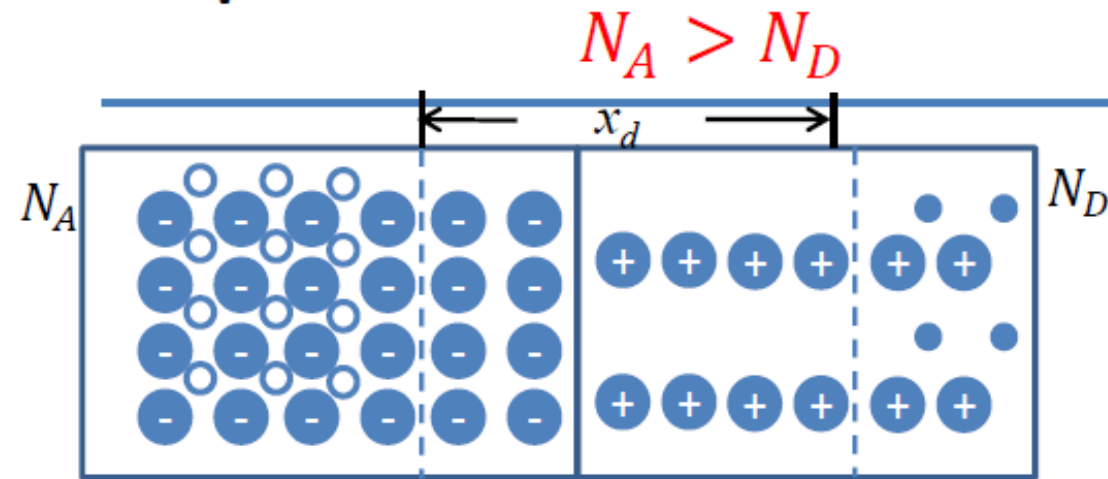
- $v_D > 0$ , **forward** bias,  $i_D > 0$ , large (good forward conduction)
- $v_D < 0$ , **reverse** bias,  $i_D < 0$ , small (poor reverse conduction)
- Goal: Find function of  $i_D$  with respect to  $v_D$ .

# p-n junction – Physical Picture



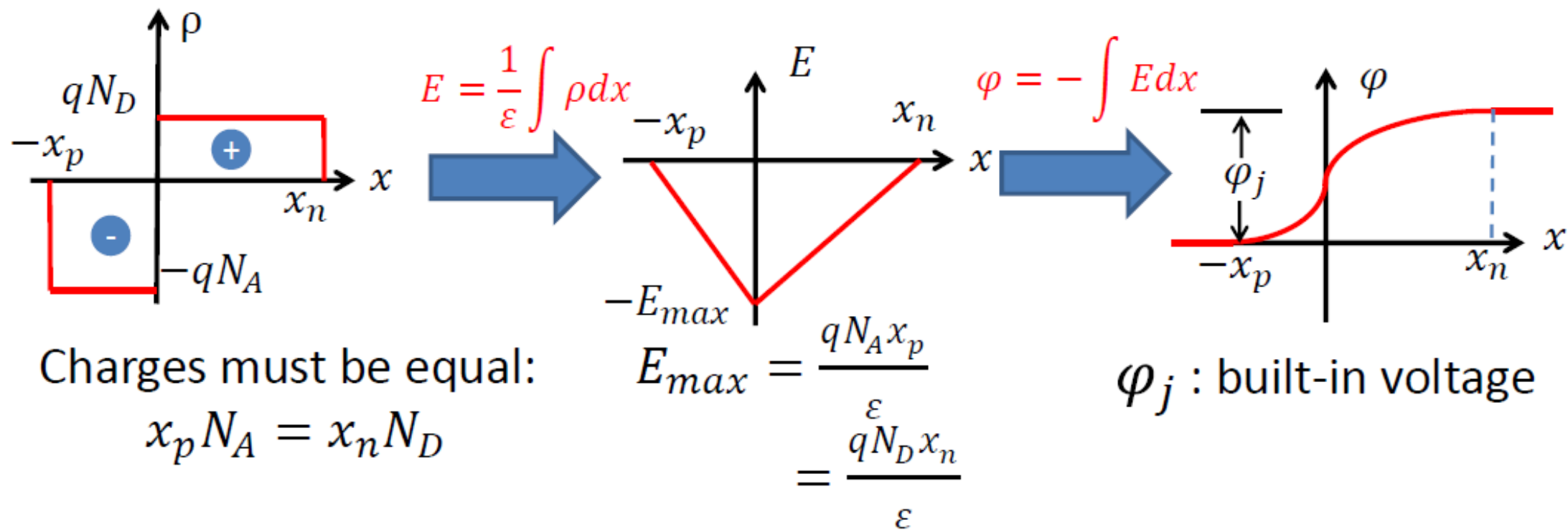
- **Diffusion** allows carriers to move from high density to low density
- Immobile ions produce a built-in electric field
- The **drift** caused by the field prevents the carriers from diffusing further.
- When drift and diffusion are balanced, a metallurgical junction forms
- The junction is featured by a **space charge region**, also called **depletion region** (*depleted* of free carriers).

# p-n Junction – Electrostatic Analysis

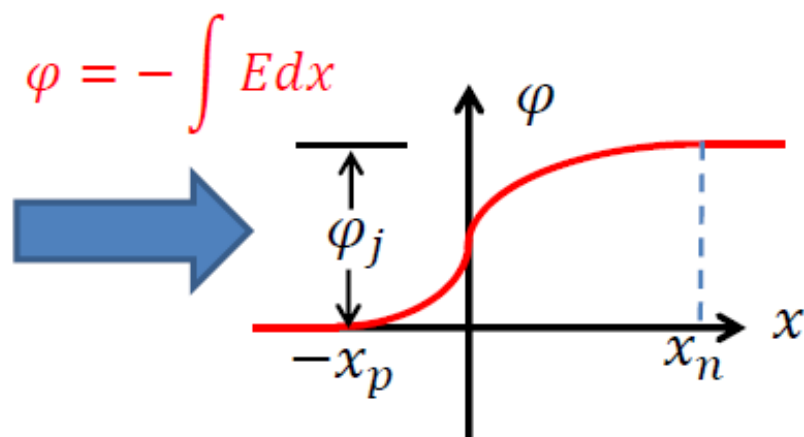
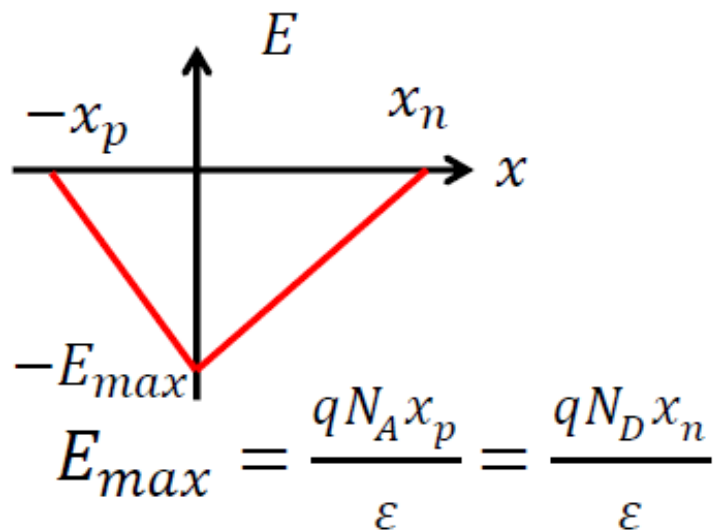


1D Poisson's Equation:

$$\frac{d^2 \phi}{dx^2} = -\frac{dE}{dx} = -\frac{\rho}{\epsilon}$$



# p-n Junction – Electrostatic Analysis



- $\varphi_j$  is the area under E-x curve:
- $\varphi_j = \frac{(x_n + x_p)}{2} |E_{max}| = \frac{qN_A(x_n + x_p)}{2\epsilon} x_p$
- Combining  $x_p N_A = x_n N_D$ , we get

$$x_p = \sqrt{\frac{2\epsilon\varphi_j}{qN_A(1+N_A/N_D)}}$$

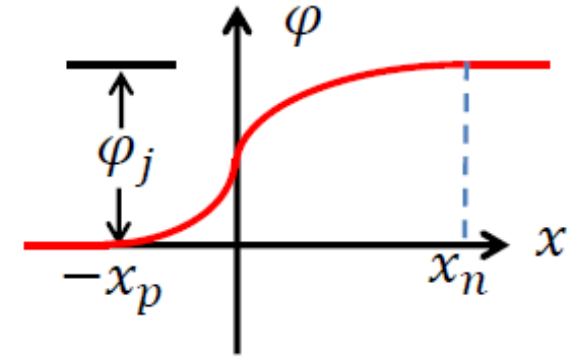
$$x_n = \sqrt{\frac{2\epsilon\varphi_j}{qN_D(1+N_D/N_A)}}$$

$$x_d = x_n + x_p = \sqrt{\frac{2\epsilon\varphi_j}{q} \left( \frac{1}{N_D} + \frac{1}{N_A} \right)}$$

Depletion width under zero bias

# p-n Junction – Electrostatic Analysis

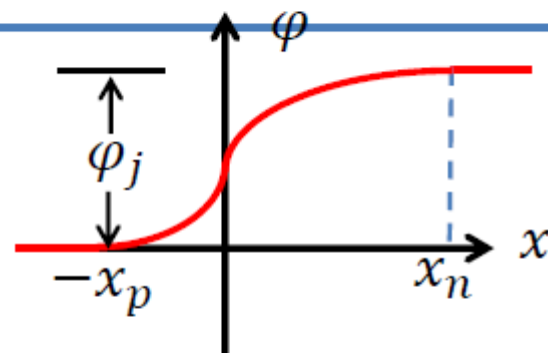
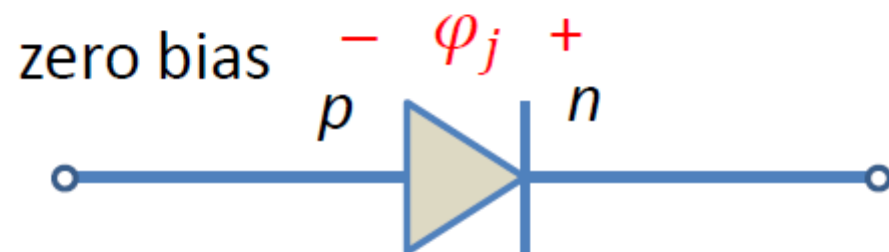
- How to find  $\varphi_j$  ?
- From detailed balance in equilibrium ( $J^{\text{drift}} = -J^{\text{diff}}$ ):



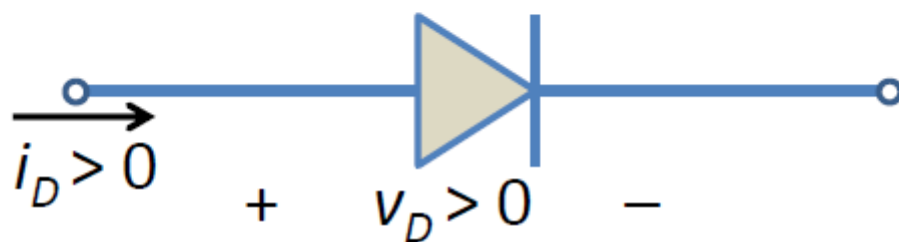
$$\varphi_j = \frac{kT}{q} \ln \frac{p_0(-x_p)}{p_0(x_n)} = \frac{kT}{q} \ln \frac{n_0(x_n)}{n_0(-x_p)} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$

- $\varphi_j$  depends on
  - Doping concentration on both sides ( $N_A, N_D$ )
  - Temperature ( $kT$ )
  - Material ( $n_i^2 \sim \exp[-E_g/kT]$ )

# p-n Junction Under Applied Voltage



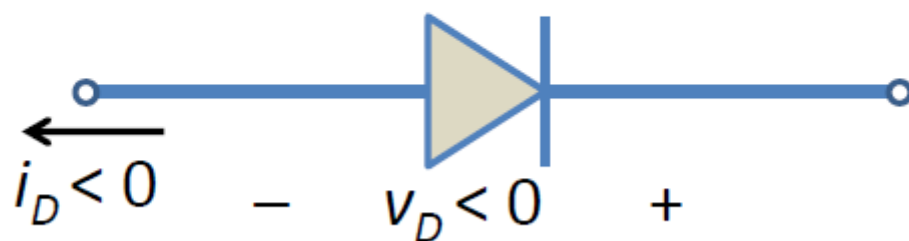
forward bias



- Under bias  $v_D$ , replace  $\phi_j$  with  $(\phi_j - V_D)$ . e.g.

$$x_d = \sqrt{\frac{2\varepsilon(\phi_j - v_D)}{q} \left( \frac{1}{N_D} + \frac{1}{N_A} \right)}$$

reverse bias



- Forward bias,  $v_D > 0$ ,  $x_d$  shrinks
- Reverse bias,  $v_D < 0$ ,  $x_d$  widens

# p-n Junction Under Applied Voltage

---

- Example: p-n junction with  $N_A = 2 \times 10^{16} \text{ cm}^{-3}$ ,  $N_D = 5 \times 10^{17} \text{ cm}^{-3}$ . Find the depletion width  $x_d$  at 300 K under (a) zero bias (b) forward bias ( $v_D = +0.5 \text{ V}$ ) (c) reverse bias ( $v_D = -5.0 \text{ V}$ ).

– At  $T=300\text{K}$ ,  $V_T = kT/q = 25.9 \text{ mV}$ ,  $n_i = 1.0 \times 10^{10} \text{ cm}^{-3}$ .

– Built-in voltage:  $\varphi_j = V_T \ln \frac{N_A N_D}{n_i^2} = 0.85 \text{ V}$

– Under zero bias:

$$x_d = \sqrt{\frac{2\varepsilon\varphi_j}{q} \left( \frac{1}{N_D} + \frac{1}{N_A} \right)}$$
$$= \sqrt{\frac{2 \cdot 11.7 \cdot 8.854 \times 10^{-14} \frac{\text{F}}{\text{cm}} \cdot 0.85 \text{ V}}{1.6 \times 10^{-19} \text{ C}} \left( \frac{1}{2 \times 10^{16} \text{ cm}^{-3}} + \frac{1}{5 \times 10^{17} \text{ cm}^{-3}} \right)} = 240 \text{ nm}$$

---



# p-n Junction Under Applied Voltage

- What happens if  $v_D > \varphi_j$  ?

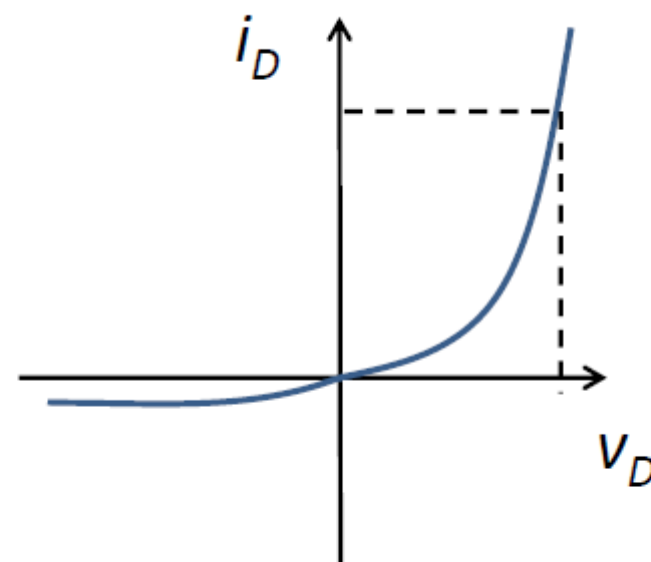
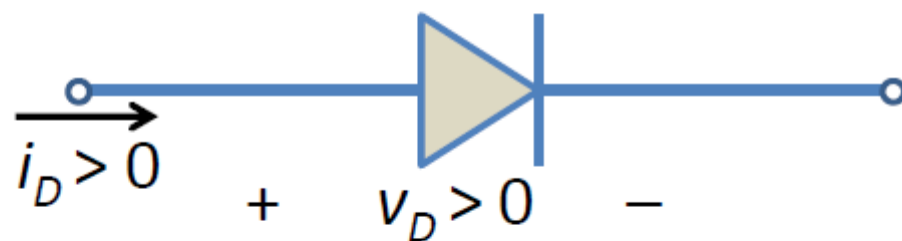
Huge currents.

Voltage drops over series R.

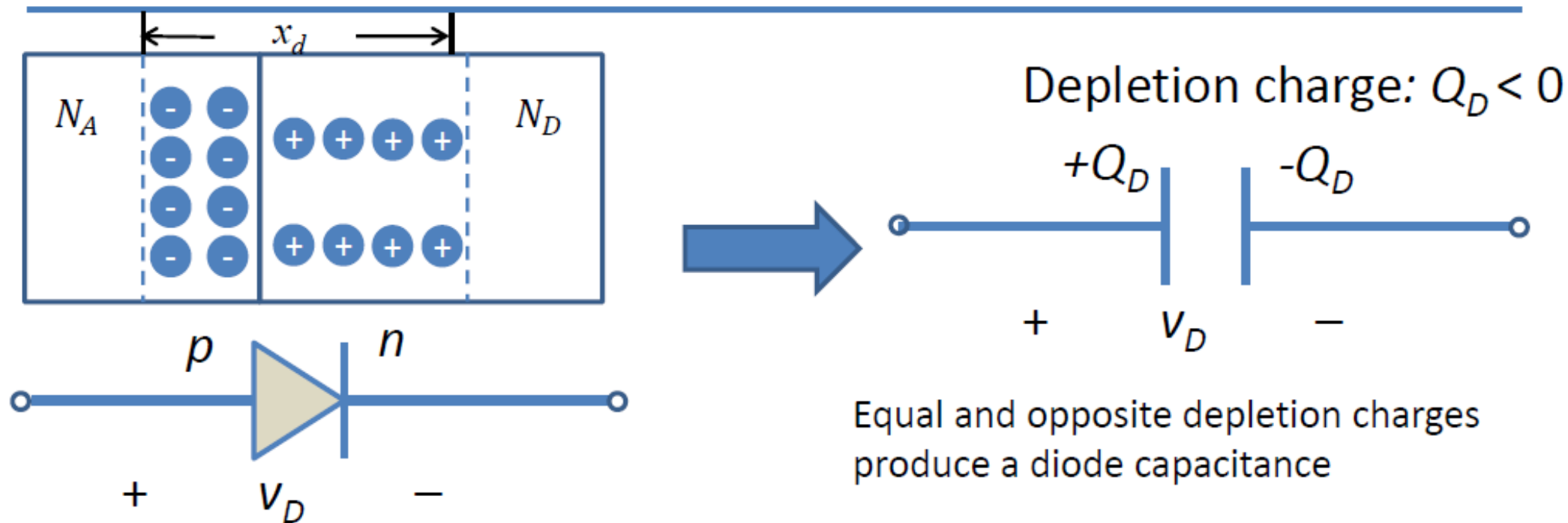
Diode destroyed before this point is reached.

$$x_d = \sqrt{\frac{2\varepsilon(\varphi_j - v_D)}{q} \left( \frac{1}{N_D} + \frac{1}{N_A} \right)}$$

forward bias



# p-n junction capacitance



- Normal linear capacitance (e.g. parallel plate):

$$C = \frac{Q}{V}$$

- “Depletion capacitance” or “junction capacitance” for diodes:

$$C_j = \frac{dQ_D}{dv_D}$$

# p-n junction capacitance

---

- Depletion capacitance:

A: diode area [cm<sup>2</sup>]

$$Q_D = -qAx_pN_A$$

$$x_p = \sqrt{\frac{2\varepsilon_s(\varphi_j - v_D)}{qN_A(1 + N_A/N_D)}}$$



$$C_j = \frac{dQ_D}{dv_D} = \frac{\varepsilon_s A}{x_d}$$

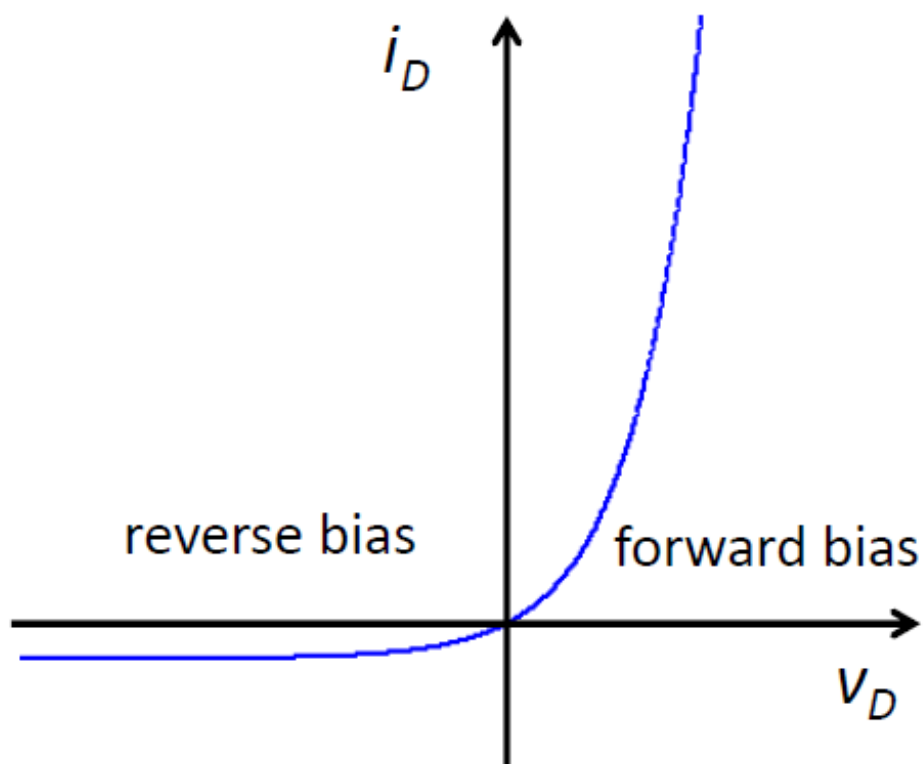
- Looks like a parallel-plate capacitor formula!
- As  $x_d$  varies with  $v_D$ ,  $C_j$  also varies with  $v_D$ .

- At zero bias,  $C_{j0} = \frac{\varepsilon_s A}{x_{d0}}$

Voltage-dependent capacitor : "varactor"

- At a non zero bias,  $C_j = \frac{\varepsilon_s A}{x_d} = C_{j0} \frac{x_{d0}}{x_d} = C_{j0} \sqrt{\frac{\varphi_j}{\varphi_j - v_D}}$

# Diode I-V Characteristics



Let  $I_S = 10^{-13} \text{ A}$ ,  $n=1$ . Then if  $i_D = 1 \text{ mA}$ ,  
 $v_d = 0.026 \ln(10^{10} + 1) = 0.60 \text{ V}$

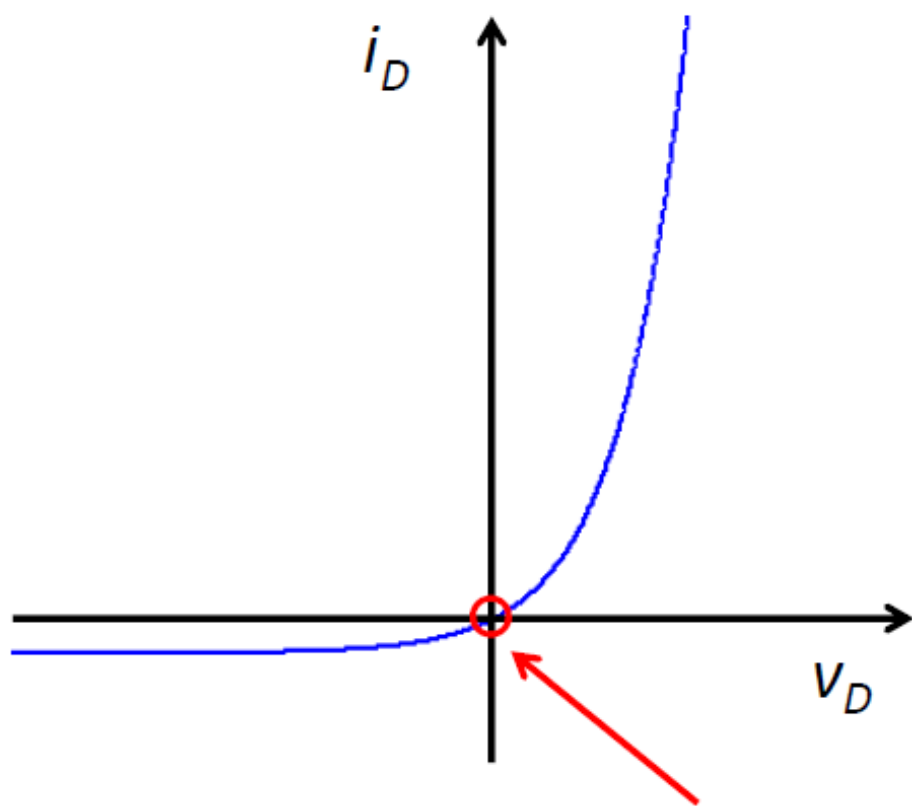
- Shockley diode equation:

$$i_D = I_S (e^{\frac{v_D}{nV_T}} - 1)$$

- $I_S$ : Reverse saturation current typically,  $10^{-18} \text{ A} \leq I_S \leq 10^{-9} \text{ A}$  (small!)
- $n$ : nonideality factor,  $n=1$  for ideal diodes (Assume  $n=1$  later on)
- $V_T$ : thermal voltage
- Can solve for  $v_d$ :

$$v_D = nV_T \ln \left( \frac{i_D}{I_S} + 1 \right)$$

# Diode I-V Zero Bias

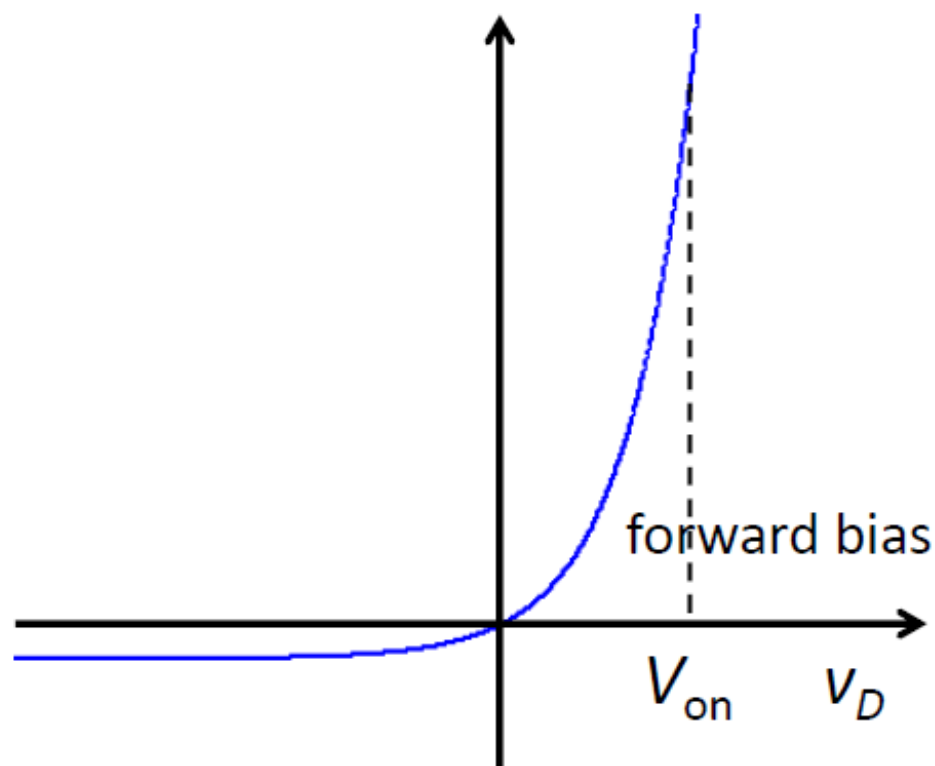


- Shockley diode equation:

$$i_D = I_S \left( e^{\frac{v_D}{V_T}} - 1 \right)$$

- Zero bias,  $v_D = 0 \Rightarrow i_D = 0$
- I-V curve passes thru origin!

# Diode I-V Forward Bias

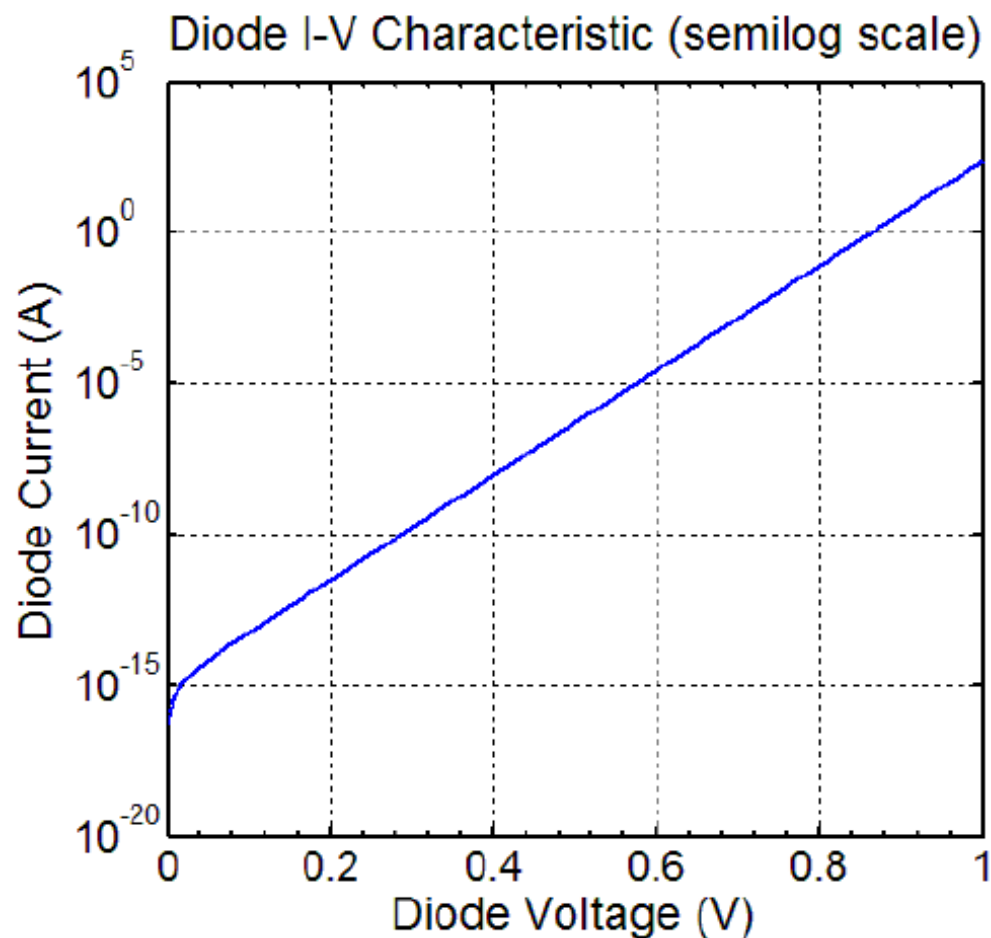


- Shockley diode equation:

$$i_D = I_S(e^{\frac{v_D}{V_T}} - 1)$$

- Forward bias,  $i_D$  large.
- If  $v_D \gg 4V_T$ , then  $e^{\frac{v_D}{V_T}} \gg 1$   
 $\Rightarrow i_D = I_S e^{\frac{v_D}{V_T}}$  (exponential)
- Turn-on voltage:** Voltage required to bring the diode into significant conduction (typically  $\sim 0.7$  V).

# Diode I-V Forward Bias



- Exponential increase

$$i_D = I_S e^{\frac{v_D}{V_T}}$$

- $\Delta V_{\text{decade}}$ : Voltage increase required to increase current by a factor of 10.

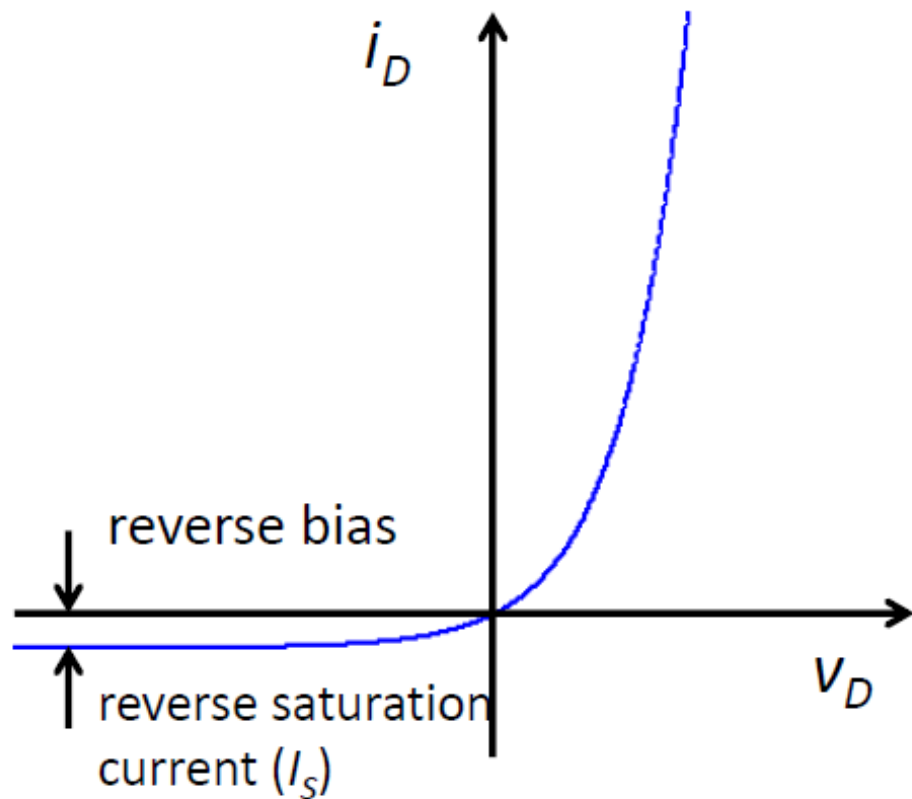
- $\Delta V_{\text{decade}} = V_T \ln 10$

- At RT,  $\Delta V_{\text{decade}} = 60 \text{ mV} \Rightarrow$

A 60 mV increase in  $v_D$  gives a **decade** increase in  $i_D$

(60n mV per decade if nonideal)

# Diode I-V Reverse Bias



- Shockley diode equation:

$$i_D = I_S(e^{\frac{v_D}{V_T}} - 1)$$

- Reverse bias,  $i_D$  small.
- If  $v_D \ll -4V_T$ , then  $e^{\frac{v_D}{V_T}} \ll 1$   
 $\Rightarrow i_D \approx -I_S$  (constant)
- $i_D$  is nearly constant under reverse bias



# Diode reverse leakage current

---

Short base  $W_p \ll L_n, W_n \ll L_p$ :

$$I_S = qAn_i^2 \left[ \frac{D_n}{N_A W_p} + \frac{D_p}{N_D W_n} \right]$$

Long base  $W_p \gg L_n, W_n \gg L_p$ :

$$I_S = qAn_i^2 \left[ \frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right]$$

Current dominated by minority carrier injection into the lightly-doped and/or narrower side.

---

# Diode reverse leakage current

Assume electron and hole diffusion length is longer than depletion width:

$$I_S = qAn_i^2 \left[ \frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right]$$

Current dominated by minority carrier injection into the lightly-doped and/or narrower side.

# Diode Circuit Analysis

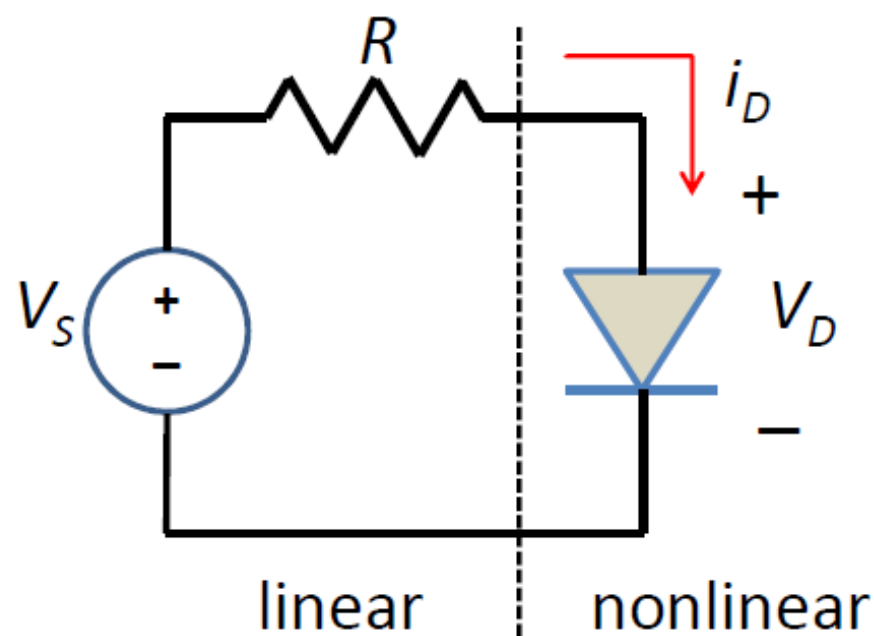
# Diode Circuit Analysis

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- Goal: find quiescent operating point (Q-point) of the diode ( $I_D$ ,  $V_D$ )
  - Analytical tools:
    - Kirchhoff's voltage law (KVL)
    - Kirchhoff's current law (KCL)
    - Element relations
  - Solution methods (Often can't solve analytically due to non-linearity)
    - Graphical methods
    - Numerical iteration
-

# Diode Circuit Analysis

- Example: Given  $v_D$ ,  $R$ , and parameters for the diode ( $I_S$ ,  $V_T$ ), find Q-point ( $I_D$ ,  $V_D$ ).



- Linear Part (resistor, load)

$$V_S = i_D R + v_D$$

- Nonlinear part (diode)

$$i_D = I_S \left( e^{\frac{v_D}{V_T}} - 1 \right)$$

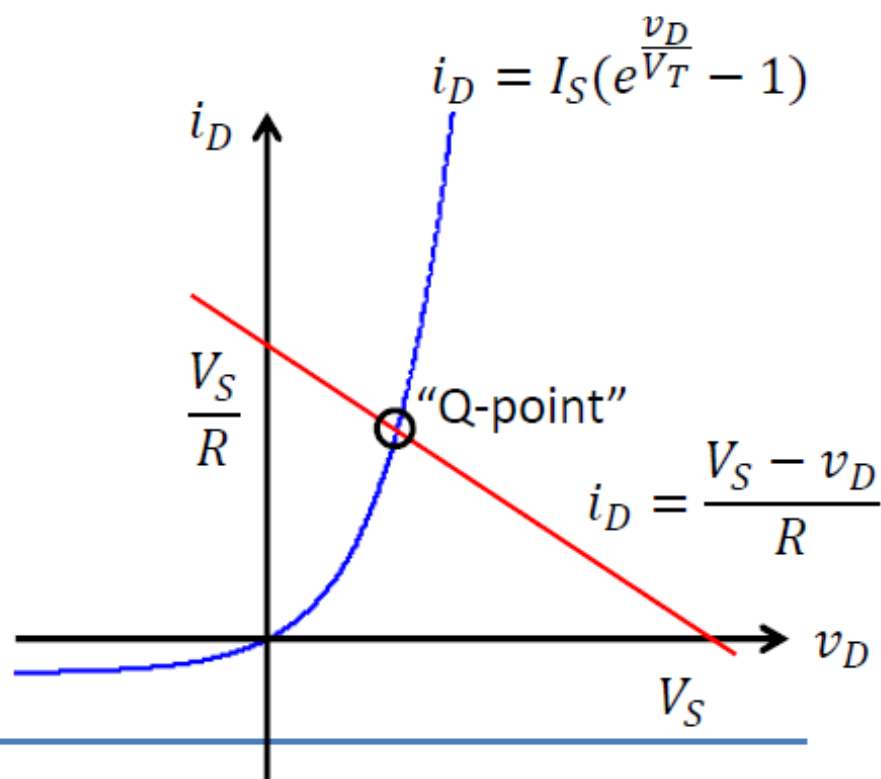
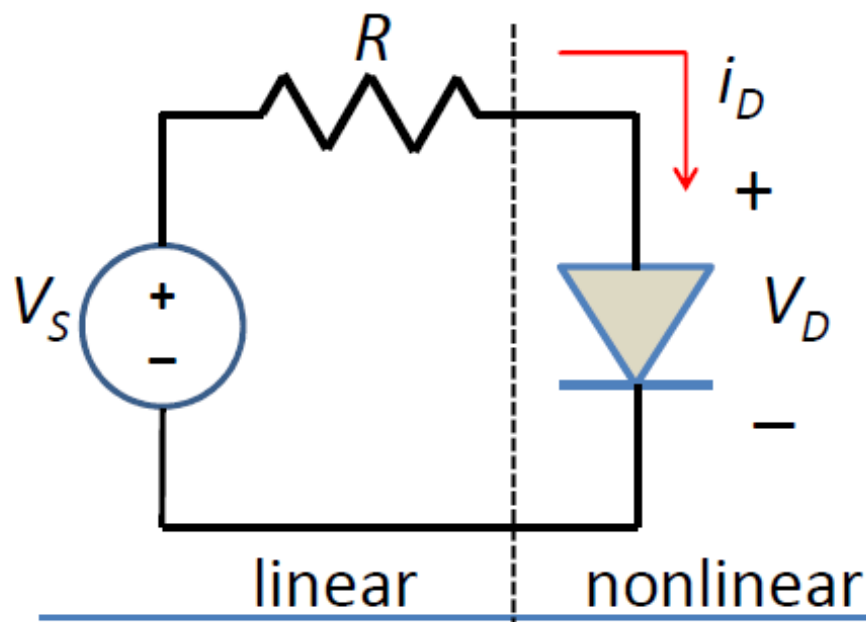
- Two equations, two unknowns  $I_D$ ,  $V_D$ .

# Load Line Analysis (Graphical)

- Write the two equations in the form of  $i_D$  v.s.  $v_D$ .

$$i_D = -\frac{v_D}{R} + \frac{V_S}{R}, \quad i_D = I_S(e^{\frac{v_D}{V_T}} - 1)$$

- Plot them on a same graph, find the intersection.



# Numerical Analysis

- Combining:

$$i_D = -\frac{v_D}{R} + \frac{V_S}{R}, \quad i_D = I_S(e^{\frac{v_D}{V_T}} - 1)$$

- We have:  $-\frac{v_D}{R} + \frac{V_S}{R} = I_S(e^{\frac{v_D}{V_T}} - 1) \Rightarrow v_D = V_T \ln\left(1 + \frac{V_S - v_D}{I_S R}\right)$
- Let  $R = 1\text{k}\Omega$ ,  $I_S = 10^{-10}\text{A}$ ,  $v_T = 26\text{ mV}$ ,  $V_S = 10\text{ V}$ . we have  
 $v_D = 0.026 \ln\left(1 + \frac{10 - v_D}{10^{-7}}\right)$
- Make an initial guess ( $v_D = 0.5\text{ V}$ ) and solve by iteration:

| Iteration #      | 0   | 1      | 2      | 3      |
|------------------|-----|--------|--------|--------|
| $v_D\text{ (V)}$ | 0.5 | 0.4776 | 0.4777 | 0.4777 |

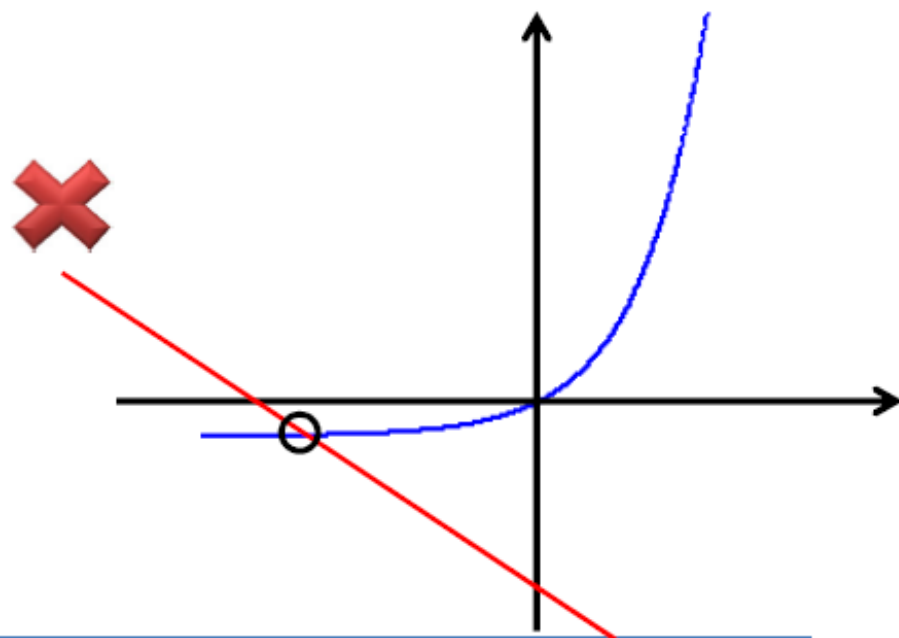
← converged !

$$i_D = -\frac{v_D}{R} + \frac{V_S}{R} = 9.522\text{ mA} \quad \text{Q-point} = (9.522\text{ mA}, 0.4777\text{ V})$$

# Numerical Analysis

- Now use . Expect diode to be reverse biased.
- From graphical analysis, . Thus
- Check for : ( — )
- Q-point =  $(10^{-10} \text{ A}, -10 \text{ V})$
- Be careful of round-off error:

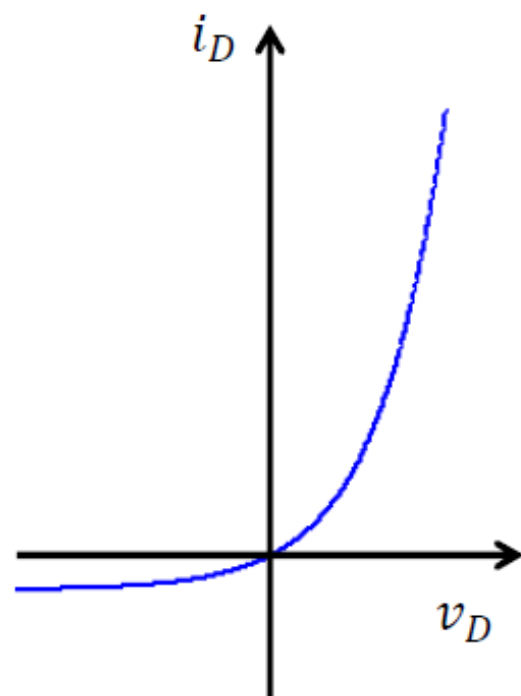
— — —  
 [ should be 9.999999999 V ]



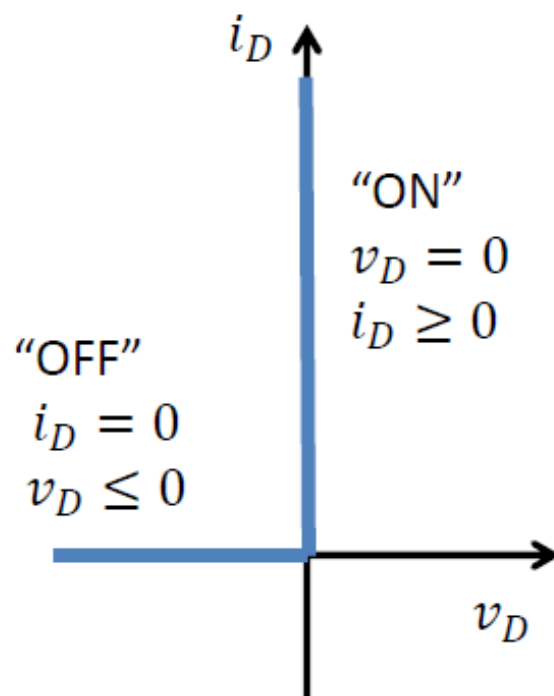


# Simplified Diode Models

General diode



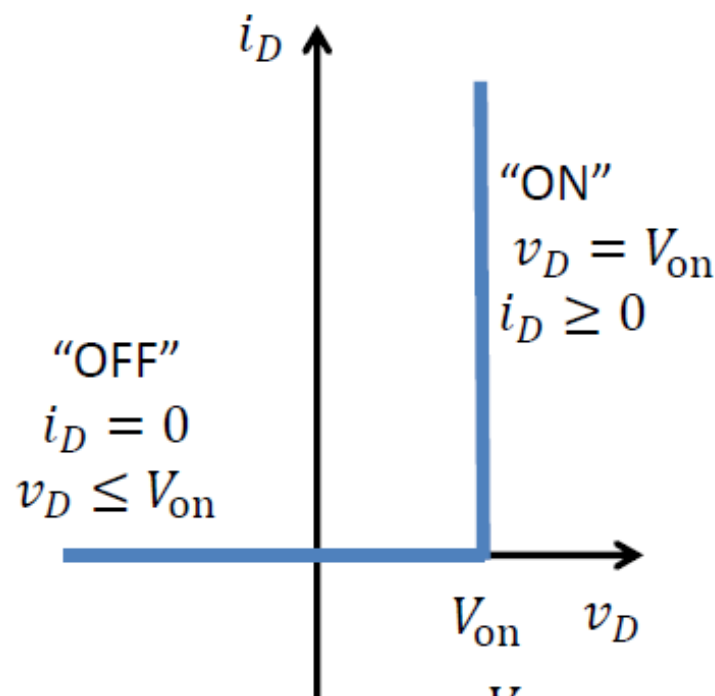
Ideal diode



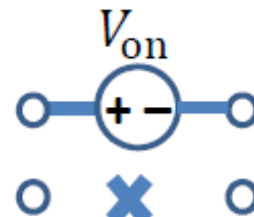
ON: short circuit  
OFF: open circuit



Constant voltage drop



ON: voltage source  
OFF: open circuit



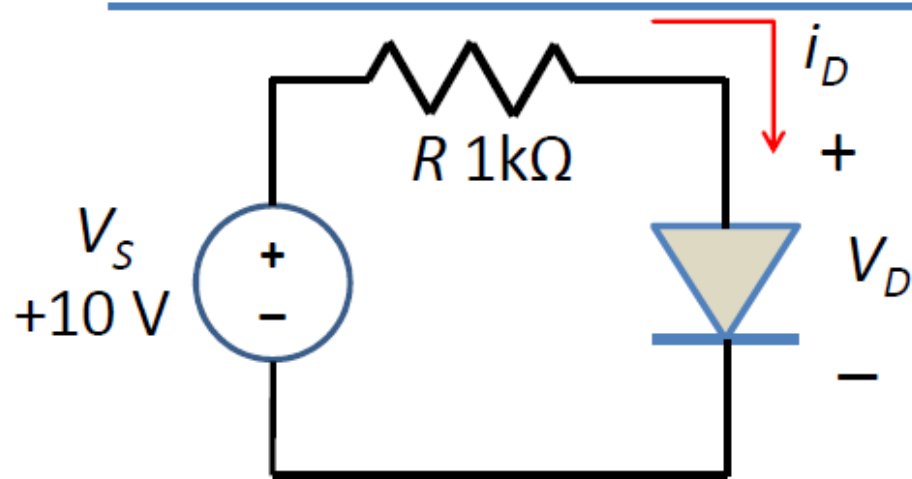
# Diode Circuit Analysis – simplified model

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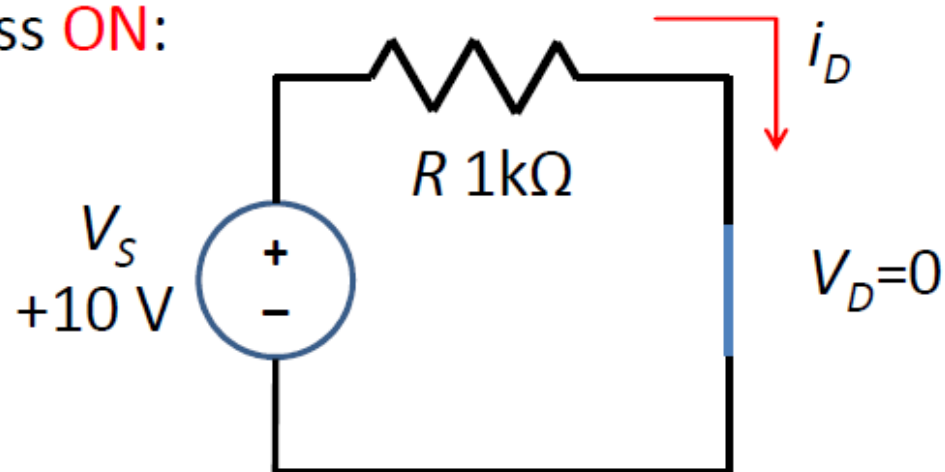
- Analysis Method:
    - Guess a state for the diode, ON or OFF
    - Replace the diode by its equivalent model for this state
    - Analyze the circuit using this equivalent model
    - Verify that the guess was correct
-

# Diode Circuit Analysis

## Ideal model



Guess **ON**:



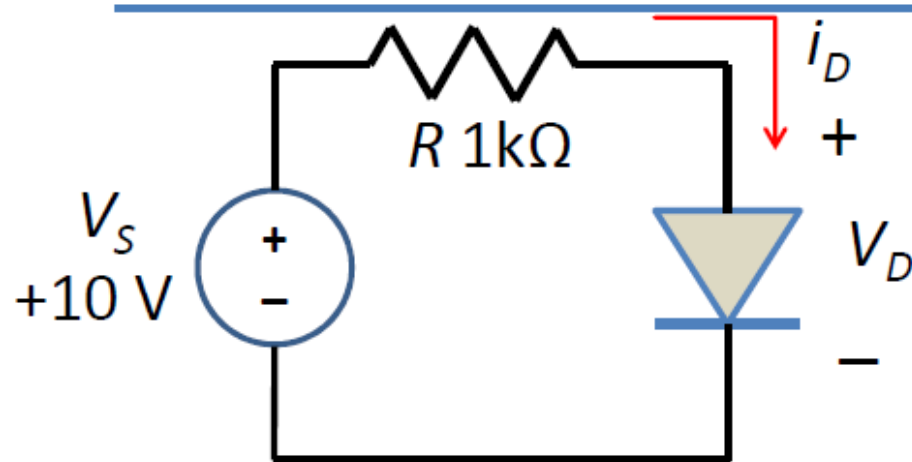
Since  $i_D = 10\text{ mA} \geq 0$ ,  $D = \text{ON}$  correct

“ON” condition:  $v_D = 0, i_D \geq 0$

---

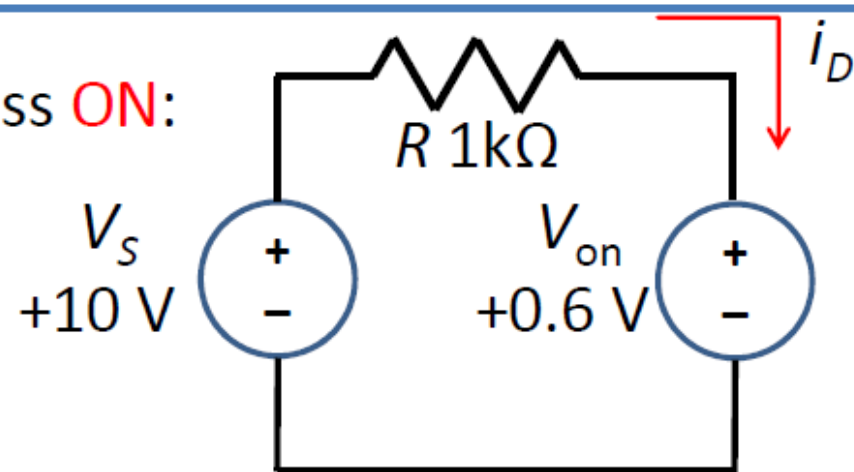
# Diode Circuit Analysis

## LVD model



$$V_{\text{on}} = 0.6\text{ V}$$

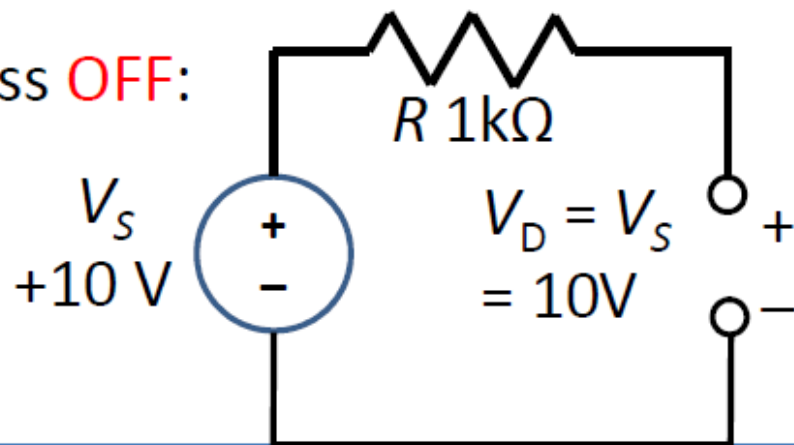
Guess **ON**:



Since  $I_D = 9.4\text{ mA} \geq 0$ , D = ON correct

“ON” condition:  $v_D = V_{\text{on}}, i_D \geq 0$

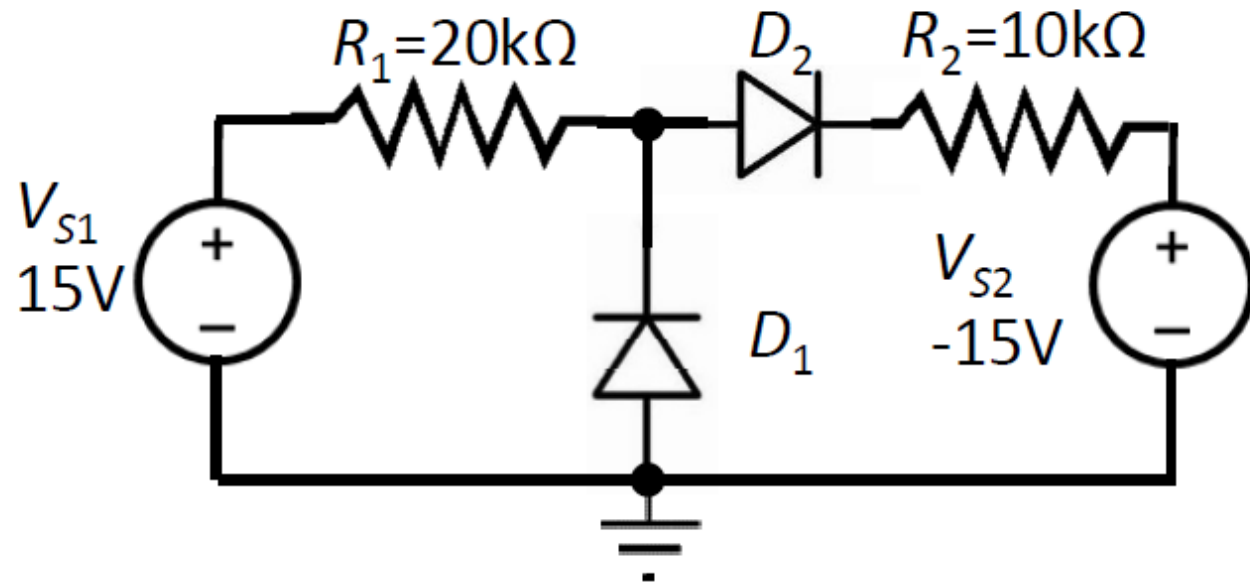
Guess **OFF**:



Since  $V_D = 10\text{ V} \geq V_{\text{on}}$ , D = OFF wrong!

“OFF” condition:  $i_D = 0, v_D \leq V_{\text{on}}$

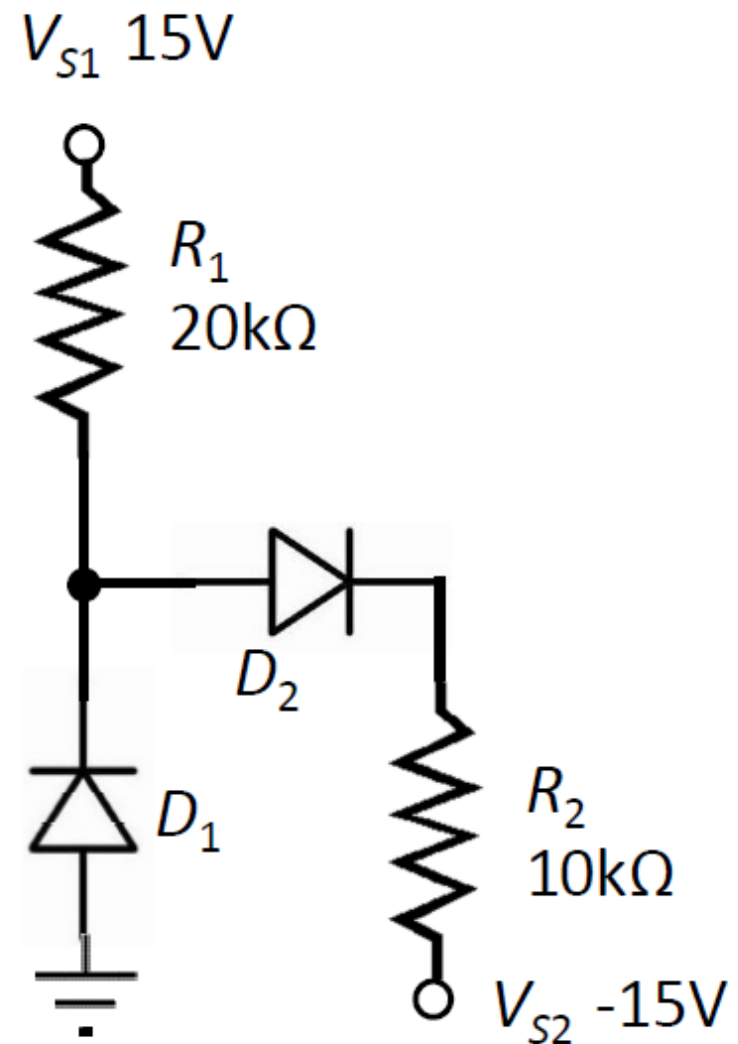
# Multiple Diode Circuits



$D_1, D_2 = \text{Ideal diodes}$

Textbook Schematic

Engineering Schematic



# Multiple Diode Circuits

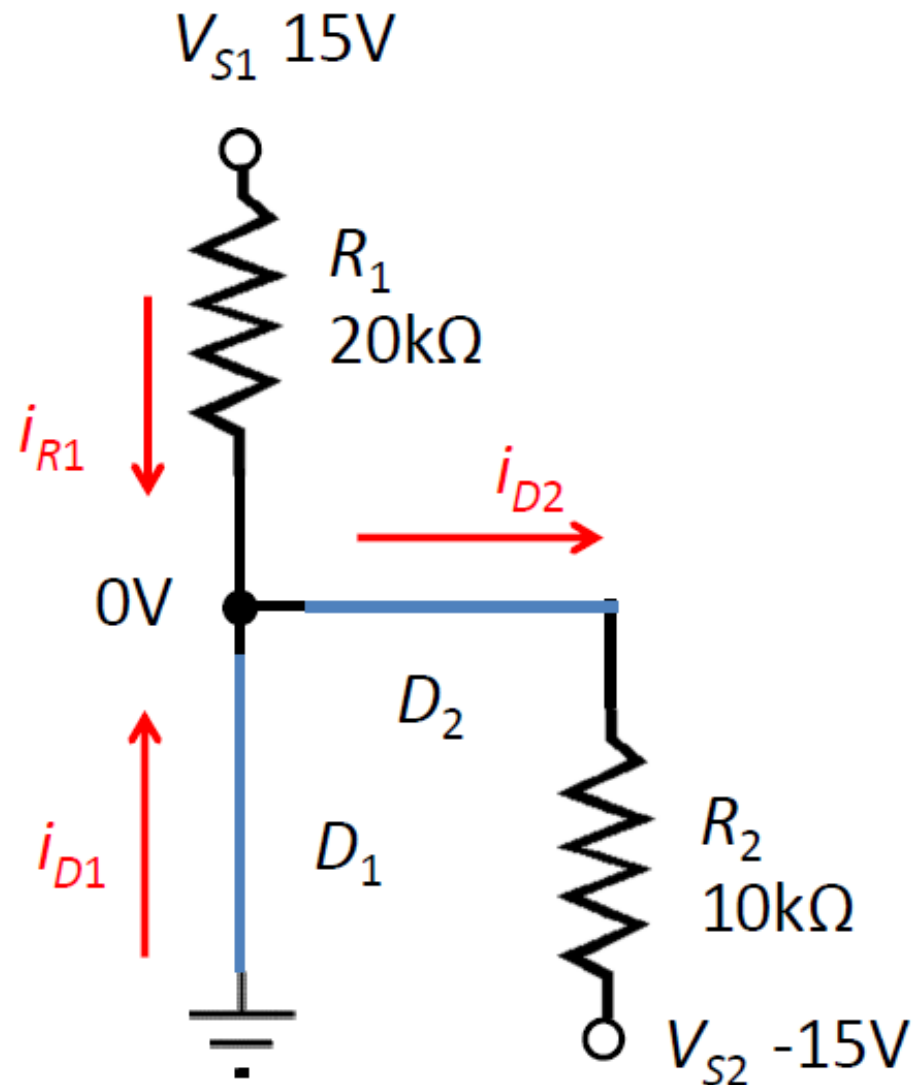
Guess:  $D_1$ ,  $D_2$  both ON

\_\_\_\_\_

\_\_\_\_\_

Check:

both  $D_1$  and  $D_2$  are ON. Correct!



# Multiple Diode Circuits

Now: Swap resistors

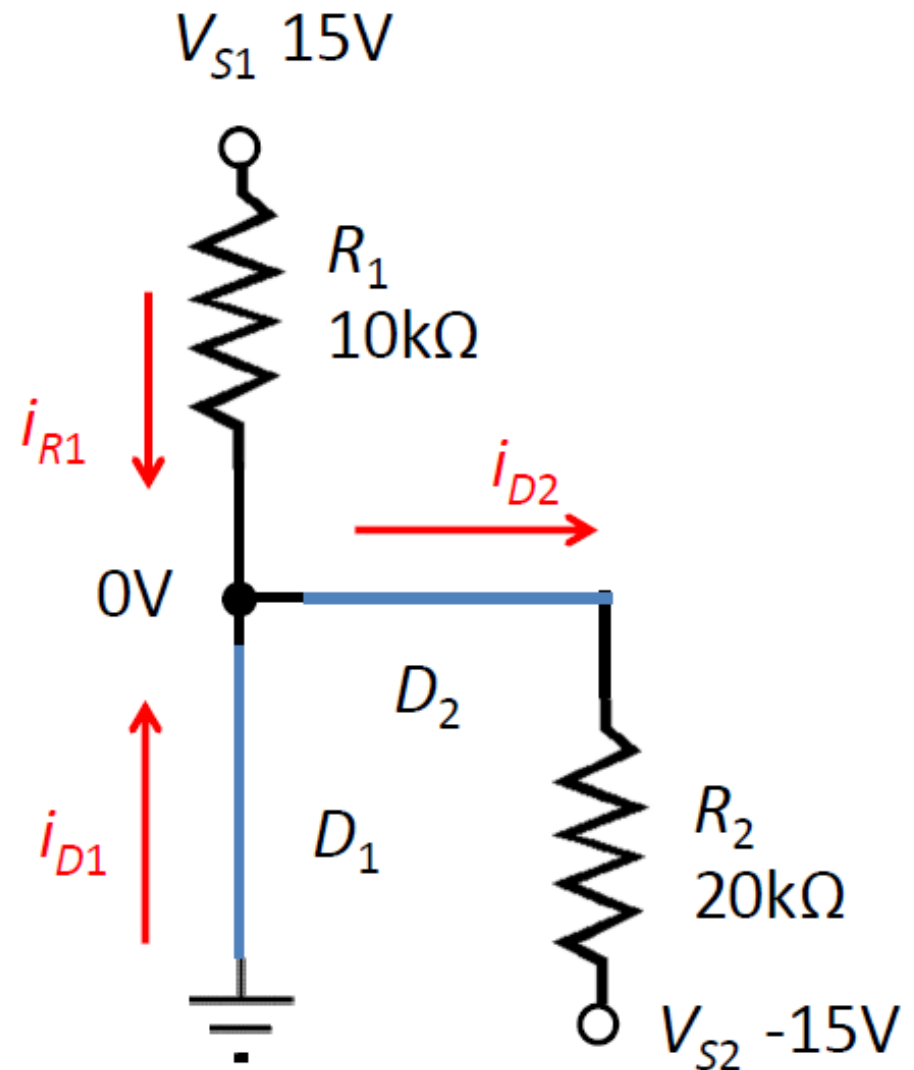
Guess:  $D_1$ ,  $D_2$  both ON

\_\_\_\_\_

\_\_\_\_\_

Check:

=> Contrary to  $D_1$  ON!



# Multiple Diode Circuits

Guess:  $D_1$  OFF,  $D_2$  ON  
                    (    )

Check:

$\Rightarrow D_1$  off. Correct.

$\Rightarrow D_2$  on. Correct.

